Linear Algebra: Homework 2

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Problem 1 [11 pts]

Label the following statements as true or false.

(a) Every vector space contains a zero vector.

(b) A vector space may have more than one zero vector.

(c) In any vector space, $ax = bx$ implies that $a = b$.

(d) In any vector space, $ax = ay$ implies that $x = y$.

(e) A vector in $\mathbb{F}^n$ may be regarded as a matrix in $M_{n\times 1}(\mathbb{F})$.

(f) An $m \times n$ matrix has m columns and n rows.

(g) In $P(\mathbb{F})$, only polynomials of the same degree may be added.

(h) If f and g are polynomials of degree n, then $f + g$ is a polynomial of degree n.

(i) If f is a polynomial of degree n and c is a nonzero scalar, then cf is a polynomial of degree n.

(j) A nonzero scalar of $\mathbb{F}$ may be considered to be a polynomial in $P(\mathbb{F})$ having degree zero.

(k) Two functions in $\mathcal{F}(S, \mathbb{F})$ are equal if and only if they have the same value at each element of S.

Answer:

The answers to these problems are, in order, (a. to e.) T, F, F, F, T, (f. to k.) F, F, F, T, T, T.

Problem 2 [2 pts]

Write the zero vector of $M_{3\times 4}(\mathbb{F})$.

Answer:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Problem 3 [4 pts]

In any vector space V, show that $(a+b)(x+y) = ax + ay + bx + by$ for any $x, y \in V$ and any $a, b \in \mathbb{F}$.

Answer:

Using axiom 8 we have $(a+b)(x+y) = a(x+y) + b(x+y)$; using axiom 7 we have $a(x+y) + b(x+y) = ax + ay + bx + by$. Therefore, $(a+b)(x+y) = ax + ay + bx + by$.

Problem 4 [4 pts]

Let V denote the set of ordered pairs of real numbers. If (a_1, a_2) and (b_1, b_2) are elements of V and $c \in \mathbb{R}$, define

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1,\ a_2 b_2) \quad \text{and} \quad c(a_1, a_2) = (ca_1,\ a_2).$$

Is V a vector space over $\mathbb{R}$ with these operations? Justify your answer.

Answer:

No, V is not a vector space over $\mathbb{R}$, since for $a, b \in \mathbb{R}$, we have $(a+b)(x, y) = ((a+b)x, y)$, while $a(x, y) + b(x, y) = (ax, y) + (bx, y) = ((a+b)x, y^2)$, which violates the axiom 8.

Problem 5 [6 pts]

Let V and W be vector spaces over a field $\mathbb{F}$. Let

$$Z = \{(v, w) : v \in V \text{ and } w \in W\}.$$

Prove that Z is a vector space over $\mathbb{F}$ with the operations

$$(v_1, w_1) + (v_2, w_2) = (v_1 + v_2,\ w_1 + w_2) \quad \text{and} \quad c(v_1, w_1) = (cv_1,\ cw_1).$$

Answer:

We prove the proposition by verifying each axiom. Let $x, y, z \in Z$ and assume $x = (v_1, w_1), y = (v_2, w_2), z = (v_3, w_3)$:

- $x + y = (v_1 + v_2, w_1 + w_2) = (v_2 + v_1, w_2 + w_1) = y + x$;
- $x + (y + z) = (v_1 + (v_2 + v_3), w_1 + (w_2 + w_3)) = ((v_1 + v_2) + v_3, (w_1 + w_2) + w_3) = (x + y) + z$;
- $0 = (0, 0), x + 0 = (v_1 + 0, w_1 + 0) = (v_1, w_1) = x$;
- $x + (-v_1, -w_1) = (v_1 - v_1, w_1 - w_1) = (0, 0) = 0$;
- $1x = (1 \cdot v_1, 1 \cdot w_1) = (v_1, w_1) = x$;
- $(ab)x = ((ab)v_1, (ab)w_1) = (a(bv_1), a(bw_1)) = a(bx)$;

Problem 5 [6 pts] continued on next page...

- $a(x+y) = (a(v_1+v_2), a(w_1+w_2)) = (av_1+av_2, aw_1+aw_2) = ax+ay$;
- $(a+b)x = ((a+b)v_1, (a+b)w_1) = (av_1+bv_1, aw_1+bw_1) = ax+bx$.

Problem 6 [7 pts]

Label the following statements as true or false.

(a) If V is a vector space and W is a subset of V that is a vector space, then W is a subspace of V.

(b) The empty set is a subspace of every vector space.

(c) If V is a vector space other than the zero vector space, then V contains a subspace W such that $W \neq V$.

(d) The intersection of any two subsets of V is a subspace of V.

(e) An $n \times n$ diagonal matrix can never have more than n nonzero entries.

(f) The trace of a square matrix is the product of its diagonal entries.

(g) Let W be the xy-plane in $\mathbb{R}^3$; that is, $W = \{(a_1, a_2, 0) : a_1, a_2 \in \mathbb{R}\}$. Then $W = \mathbb{R}^2$.

Answer:

The answers to these problems are, in order, F, F, T, F, T, F, F.

Problem 7 [6 pts]

Determine whether the following sets are subspaces of $\mathbb{R}^3$ under the operations of addition and scalar multiplication defined on $\mathbb{R}^3$. Justify your answers.

(a) $W_1 = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 = 3a_2 \text{ and } a_3 = -a_2\}$

(b) $W_2 = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 = a_3 + 2\}$

Answer:

(a) W_1 is a subspace of $\mathbb{R}^3$. Let $x = (3v_1, v_1, -v_1), y = (3v_2, v_2, -v_2)$,

- $(0,0,0)$ satisfies $a_1 = 3a_2$ and $a_3 = -a_2$;
- $x+y = (3(v_1+v_2), v_1+v_2, -(v_1+v_2))$ is in W_1;
- $ax = (3av_1, av_1, -av_1)$ is in W_1.

(b) W_2 is not a subspace of $\mathbb{R}^3$, since $(0,0,0)$ does not satisfy $a_1 = a_3 + 2$ and is not in W_2.

Problem 8 [6 pts]

Prove that $W_1 = \{(a_1, a_2, \ldots, a_n) \in \mathbb{F}^n : a_1 + a_2 + \cdots + a_n = 0\}$ is a subspace of $\mathbb{F}^n$, but $W_2 = \{(a_1, a_2, \ldots, a_n) \in \mathbb{F}^n : a_1 + a_2 + \cdots + a_n = 1\}$ is not.

Answer:

For W_1, the 0 vector is in W_1, any addition and scalar multiplication in W_1 keep the 0-sum property, which implies it is closed under addition and scalar multiplication. Therefore, W_1 is a subspace of $\mathbb{F}^n$.

However, the 0 vector is not in W_2, so W_2 is not a subspace.

Problem 9 [10 pts]

Let W_1 and W_2 be subspaces of a vector space V. Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Answer:

$\Longrightarrow$: Assume for contradiction that $W_1 \cup W_2$ is a subspace, while $W_1 \nsubseteq W_2$ and $W_2 \nsubseteq W_1$, then we find x, y which only belong to W_1 and W_2, respectively. We have $x + y \in W_1 \cup W_2$ and assume w.l.o.g. that $x + y \in W_1$, then we have $z \in W_1$ such that $x + y + z = 0$. Therefore, $x + z = -y \in W_1$, and hence $y \in W_1$, which violates the assumption that $y \notin W_1$.

$\Longleftarrow$: On the other hand, it is clear that if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$, then $W_1 \cup W_2$ is a subspace.

In conclusion, we have proven that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Definition 1

If S_1 and S_2 are nonempty subsets of a vector space V, then the **sum** of S_1 and S_2, denoted $S_1 + S_2$, is the set $\{x + y : x \in S_1 \text{ and } y \in S_2\}$.

Definition 2

A vector space V is called the **direct sum** of W_1 and W_2 if W_1 and W_2 are subspaces of V such that $W_1 \cap W_2 = \{0\}$ and $W_1 + W_2 = V$. We denote that V is the direct sum of W_1 and W_2 by writing $V = W_1 \oplus W_2$.

Problem 10 [10 pts]

Let W_1 and W_2 be subspaces of a vector space V.

(a) Prove that $W_1 + W_2$ is a subspace of V that contains both W_1 and W_2.

(b) Prove that any subspace of V that contains both W_1 and W_2 must also contain $W_1 + W_2$.

Problem 10 [10 pts] continued on next page…

Answer:

(a)
- $0 \in W_1 + W_2$ since $0 \in W_1$ and $0 \in W_2$;
- For $x, y \in W_1 + W_2$, let $x = x_1 + x_2$ and $y = y_1 + y_2$ with $x_1, y_1 \in W_1, x_2, y_2 \in W_2$ by the definition, we have $x_1 + y_1 \in W_1, x_2 + y_2 \in W_2$, then $x + y = (x_1 + y_1) + (x_2 + y_2)$ is in $W_1 + W_2$.
- For $x \in W_1 + W_2$, let $x = x_1 + x_2$ with $x_1 \in W_1, x_2 \in W_2$ by the definition, we have $ax_1 \in W_1, ax_2 \in W_2$, then $ax = a(x_1 + x_2) = ax_1 + ax_2$ is in $W_1 + W_2$.
- Since $0 \in W_1$ and $0 \in W_2$, for any $x \in W_1$ and $y \in W_2$, we have $x + 0 \in W_1 + W_2$ and $0 + y \in W_1 + W_2$.

(b) Assume for contradiction that subspace W contains both W_1 and W_2, while $x \in W_1 + W_2$ is not in W. Let $x = x_1 + x_2$ with $x_1 \in W_1$ and $x_2 \in W_2$, then $x_1, x_2 \in W$, so $x_1 + x_2 = x \in W$, contradicting the assumption.

Therefore, any subspace of V that contains both W_1 and W_2 must also contain $W_1 + W_2$.

Problem 11 [20 pts]

Let W be a subspace of a vector space V over a field $\mathbb{F}$. For any $v \in V$ the set $\{v\} + W = \{v + w : w \in W\}$ is called the **coset** of W containing v. It is customary to denote this coset by $v + W$ rather than $\{v\} + W$.

(a) Prove that $v + W$ is a subspace of V if and only if $v \in W$.

(b) Prove that $v_1 + W = v_2 + W$ if and only if $v_1 - v_2 \in W$.

Addition and scalar multiplication by scalars of $\mathbb{F}$ can be defined in the collection $S = \{v + W : v \in V\}$ of all cosets of W as follows:

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$$

for all $v_1, v_2 \in V$ and

$$a(v + W) = av + W$$

for all $v \in V$ and $a \in \mathbb{F}$.

(c) Prove that the preceding operations are well defined; that is, show that if $v_1 + W = v_1' + W$ and $v_2 + W = v_2' + W$, then

$$(v_1 + W) + (v_2 + W) = (v_1' + W) + (v_2' + W)$$

and

$$a(v_1 + W) = a(v_1' + W)$$

for all $a \in \mathbb{F}$.

Problem 11 [20 pts] continued on next page…

(d) Prove that the set S is a vector space with the operations defined in (c). This vector space is called the **quotient space of V modulo W** and is denoted by V/W.

Answer:

(a) $\Longrightarrow$: since $v+W$ is a subspace, then $0 \in v+W$, implies $-v \in W$ thus $v \in W$;

$\Longleftarrow$: since $v \in W$, then $-v \in W$, thus $0 = -v+v \in v+W$; for $x, y \in v+W$, we have $x+y = v+v+(x-v)+(y-v)$, since $v, x-v, y-v \in W$, we have $v+(x-v)+(y-v) \in W$ then $x+y \in v+W$; for $x \in v+W$, we have $ax = v+(a-1)v+a(x-v)$, since $v, x-v \in W$, we have $(a-1)v+a(x-v) \in W$ then $ax \in v+W$.

(b) $\Longrightarrow$: since $0 \in W$, we have $v_1 \in v_1+W = v_2+W$, then $v_1 - v_2 \in W$.

$\Longleftarrow$: since $v_1 - v_2 \in W$, then for any $x \in v_1+W$, we have $x = v_2+(x-v_2) = v_2+(x-v_1+v_1-v_2)$; since $x-v_1, v_1-v_2 \in W$, we have $x-v_1+v_1-v_2 \in W$ then $x \in v_2+W$; similarly, we have $v_2-v_1 \in W$ and derive the opposite direction.

(c) Since $v_1+W = v_1'+W$ and $v_2+W = v_2'+W$, we obtain $v_1-v_1', v_1'-v_1, v_2-v_2', v_2'-v_2 \in W$. For any $x \in (v_1+W)+(v_2+W)$, we write it into $x = v_1+x_1+v_2+x_2 = v_1'+(v_1-v_1'+x_1)+v_2'+(v_2-v_2'+x_2)$, thus $x \in (v_1'+W)+(v_2'+W)$. Similar for the opposite direction.

For $x \in a(v_1+W)$, we write $x = a(v_1+x_1) = a(v_1'+(v_1-v_1'+x_1)) \in a(v_1'+W)$. Similar for the opposite direction.

(d)

- $(v_1+W)+(v_2+W) = (v_1+v_2)+W = (v_2+v_1)+W = (v_2+W)+(v_1+W)$;
- $((v_1+W)+(v_2+W))+(v_3+W) = (v_1+v_2+v_3)+W = (v_1+W)+((v_2+W)+(v_3+W))$;
- $0 = (0+W)$, $(v_1+W)+(0+W) = (v_1+0)+W = v_1+W$;
- For any (v_1+W), $(v_1+W)+(-v_1+W) = (v_1-v_1)+W = 0$;
- $1 \cdot (v_1+W) = 1 \cdot v_1 + W = v_1+W$;
- $(ab) \cdot (v_1+W) = (ab)v_1+W = a(b(v_1+W))$;
- $a((v_1+W)+(v_2+W)) = a((v_1+v_2)+W) = a(v_1+v_2)+W = a(v_1+W)+a(v_2+W)$;
- $(a+b)(v_1+W) = (a+b)v_1+W = a(v_1+W)+b(v_1+W)$.

Problem 12 [6 pts]

Label the following statements as true or false.

(a) If S is a linearly dependent set, then each vector in S is a linear combination of other vectors in S.

(b) Any set containing the zero vector is linearly dependent.

(c) The empty set is linearly dependent.

(d) Subsets of linearly dependent sets are linearly dependent.

(e) Subsets of linearly independent sets are linearly independent.

(f) If $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$ and $x_1, x_2, \ldots, x_n$ are linearly independent, then all the scalars a_i are zero.

Problem 12 [6 pts] continued on next page...

Answer:

The answers to these problems are, in order, F, T, F, F, T, T.

Problem 13 [6 pts]

Determine whether the following sets are linearly dependent or linearly independent.

(a) $\left\{\begin{pmatrix} 1 & -3 \\ -2 & 4 \end{pmatrix}, \begin{pmatrix} -2 & 6 \\ 4 & -8 \end{pmatrix}\right\}$ in $M_{2\times 2}(\mathbb{R})$

(b) $\{x^3 + 2x^2,\ -x^2 + 3x + 1,\ x^3 - x^2 + 2x - 1\}$ in $P_3(\mathbb{R})$

Answer:

(a) Let A_1, A_2 be the two matrices, then $2A_1 + A_2 = 0$, here the 0 refers to the zero of $M_{2\times 2}(\mathbb{R})$. This implies $S = \{A_1, A_2\}$ is not linearly independent.

(b) This set $S = \{f_1, f_2, f_3\}$ is linearly independent. Assume $x_1 f_1 + x_2 f_2 + x_3 f_3 = 0$, then we have

$$\begin{aligned} x_1 \qquad\quad &+x_3 = 0, \\ 2x_1 \ -x_2 \ &-x_3 = 0, \\ +3x_2 \ &+2x_3 = 0, \\ +x_2 \ &-x_3 = 0. \end{aligned}$$

Since these equations have only the zero solution, the set is linearly independent.

Problem 14 [6 pts]

In $M_{m\times n}(\mathbb{F})$, let E^{ij} denote the matrix whose only nonzero entry is 1 in the ith row and jth column. Prove that $\{E^{ij} : 1 \le i \le m,\ 1 \le j \le n\}$ is linearly independent.

Answer:

The coefficient of E^{ij} only affects the result on position (i, j), thus if every position of the summation matrix is 0, then all the coefficients are 0.

Therefore, $\{E^{ij}\}$ is linearly independent.

Problem 15 [10 pts]

Let $S = \{u_1, u_2, \ldots, u_n\}$ be a linearly independent subset of a vector space V over the field $\mathbb{Z}_2$. How many vectors are there in Span(S)? Justify your answer.

Answer:

There are 2^n vectors in Span(S); these vectors are $a_1u_1 + a_2u_2 + \cdots + a_nu_n$, as $a_i \in \{0, 1\}$, and different choices of the coefficients give different vectors. Indeed, if $a_1u_1 + \cdots + a_nu_n = b_1u_1 + \cdots + b_nu_n$, then $(a_1 - b_1)u_1 + \cdots + (a_n - b_n)u_n = 0$. Since S is linearly independent, we have $a_i - b_i = 0$.

Problem 16 [10 pts]

Prove that if $\{A_1, A_2, \ldots, A_k\}$ is a linearly independent subset of $M_{n\times n}(\mathbb{F})$, then $\{A_1^\top, A_2^\top, \ldots, A_k^\top\}$ is also linearly independent.

Answer:

Assume for contradiction that $a_1A_1^\top + \cdots + a_kA_k^\top = 0$ with some $a_i \neq 0$. Since the transpose only changes positions but not values, the same coefficients give $a_1A_1 + \cdots + a_kA_k = 0$, which contradicts the linear independence of $\{A_1, \ldots, A_k\}$.

Therefore, $\{A_1^\top, \ldots, A_k^\top\}$ is linearly independent.

Problem 17 [10 pts]

Let $f, g \in \mathcal{F}(\mathbb{R}, \mathbb{R})$ be the functions defined by $f(t) = e^{rt}$ and $g(t) = e^{st}$, where $r \neq s$. Prove that f and g are linearly independent in $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Answer:

Assume for contradiction that $af + bg = 0$. We consider two points $t = 0, t = 1$, which give $a + b = 0$ and $ae^r + be^s = 0$, and this system has only the zero solution.

Therefore, f and g are linearly independent in $\mathcal{F}(\mathbb{R}, \mathbb{R})$.

Problem 18 [6 pts]

Determine which of the following sets are bases for $\mathbb{R}^3$.

(a) $\{(1, 0, -1),\ (2, 5, 1),\ (0, -4, 3)\}$

(b) $\{(2, -4, 1),\ (0, 3, -1),\ (6, 0, -1)\}$

Answer:

(a) We determine it with the augmented matrix:

$$\begin{bmatrix} 1 & 2 & 0 & * \\ 0 & 5 & -4 & * \\ -1 & 1 & 3 & * \end{bmatrix},$$

and reduce it into reduced echelon form:

$$\begin{bmatrix} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{bmatrix}.$$

Since the last column is not a pivot column, any $\boldsymbol{b} \in \mathbb{R}^3$ is in $\text{Span}\{(1,0,-1),\ (2,5,1),\ (0,-4,3)\}$, and the set is a basis of $\mathbb{R}^3$.

(b) We determine it with the augmented matrix:

$$\begin{bmatrix} 2 & 0 & 6 & * \\ -4 & 3 & 0 & * \\ 1 & -1 & -1 & * \end{bmatrix},$$

and reduce it into reduced echelon form:

$$\begin{bmatrix} 1 & 0 & 3 & * \\ 0 & 1 & 4 & * \\ 0 & 0 & 0 & * \end{bmatrix}.$$

Since the coefficient matrix has only two pivot columns (its reduced echelon form has a zero row), there are vectors $\boldsymbol{b} \in \mathbb{R}^3$ that are not in $\text{Span}\{(2,-4,1),\ (0,3,-1),\ (6,0,-1)\}$, and the set is not a basis of $\mathbb{R}^3$.

Problem 19 [4 pts]

Is $\{(1,4,-6),\ (1,5,8),\ (2,1,1),\ (0,1,0)\}$ a linearly independent subset of $\mathbb{R}^3$? Justify your answer.

Answer:

There are 4 vectors in the set, while the dimension of $\mathbb{R}^3$ is only 3 ($\{(1,0,0),(0,1,0),(0,0,1)\}$ is a basis of $\mathbb{R}^3$). This implies that the set $\{(1,4,-6),\ (1,5,8),\ (2,1,1),\ (0,1,0)\}$ is linearly dependent.

Problem 20 [16 pts]

Consider a system of linear equations

$$\begin{aligned} (a_1+b)x_1 &+a_2x_2 &+a_3x_3 &+\cdots &+a_nx_n &= 0 \\ a_1x_1 &+(a_2+b)x_2 &+a_3x_3 &+\cdots &+a_nx_n &= 0 \\ a_1x_1 &+a_2x_2 &+(a_3+b)x_3 &+\cdots &+a_nx_n &= 0 \\ & & & \cdots & & \\ a_1x_1 &+a_2x_2 &+a_3x_3 &+\cdots &+(a_n+b)x_n &= 0 \end{aligned}$$

where $\sum_{i=1}^{n} a_i \neq 0$.

Problem 20 [16 pts] continued on next page…

Determine the relationship between $a_1, a_2, \cdots, a_n$ and b such that

(a) This linear system has only a zero solution. (8 pts)

(b) This system has nonzero solutions. In this case, determine a solution set. (8 pts)

Answer:

We can first obtain that $bx_1 = bx_2 = \cdots = bx_n$, thus $x_1 = x_2 = \cdots = x_n$ or $b = 0$.

If $b = 0$, then the equations are equivalent to $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$.

If $b \neq 0$, then $x_1 = x_2 = \cdots = x_n = x$ is required, and the equations are equivalent to $(\sum_{i=1}^{n} a_i + b)x = 0$.

(a) If $b = 0$, then $n = 1$ is required for the system to have only the zero solution, since there is only one constraint ($a_1 = \sum_{i=1}^{n} a_i \neq 0$ is guaranteed).

Otherwise, if $b \neq 0$, the equation $(\sum_{i=1}^{n} a_i + b)x = 0$ has only the zero solution if and only if $b + \sum_{i=1}^{n} a_i \neq 0$.

(b) If $b = 0$, then $n \neq 1$ will lead to nonzero solutions. In this case, we can find any $a_i \neq 0$, let $x_j = 1$ for all $j \neq i$, and use x_i to adjust the sum to zero. The solution set is $a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0$.

If $b \neq 0$, then $b + \sum_{i=1}^{n} a_i = 0$ will lead to nonzero solutions. The solution set is $x_1 = x_2 = \cdots = x_n$.